PackForge · Falak Anwar · Zaid Alam

PackForge shows that a 48 V scooter pack charges hidden cells at 47.6 °C after a hot ride, and ranks three fixes before any hardware is built

The problem
Problem understanding · Desirability

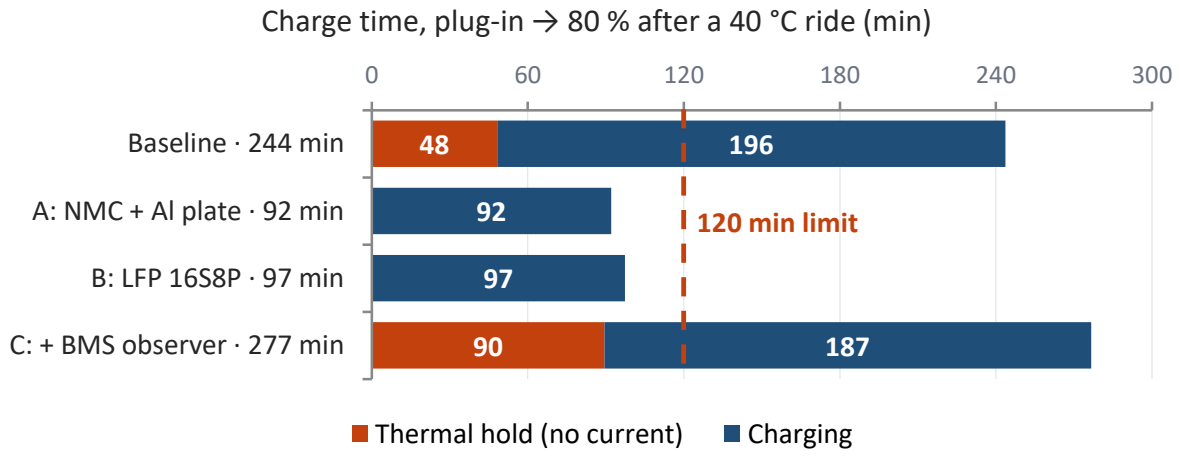
- 48 V e-scooter pack (13S10P NMC, 2,263 Wh) at 40 °C, plugged in straight after the ride
- Edge NTCs read 3.2 K low; after a 48 min hold the BMS charges interior cells at 47.6 °C (limit 45 °C)
- For students learning why designs fail and pack engineers choosing what to prototype

The platform
Feasibility · Innovation

- One design file feeds an electro-thermal sim, a sensor-only BMS and a conflict checker (11 requirements, 14 interface rules)
- New: cause chains with fixes, blind-spot metric, virtual hotspot sensor, evidence levels
- 31 s per study on a laptop; a new option is one JSON file

The evidence
Engineering quality · Prototype

- Cell fitted on public TU Berlin data (proxy cell LG M50LT) at 25 and 35 °C
- 60 hold-out logs at 45 °C: voltage RMSE 17–18 mV (median)
- Heating underestimated ~2×, so every design is re-run at 2× heat
- 23 automated tests incl. heat balance and a hand-calc bound



Recommendation
Business viability

- **B · LFP for new packs:** 97 min, ₹18.2 k per usable kWh (lowest), ~3× cycle life (estimate); over mass and volume targets
- **A · NMC + Al plate if mass is capped:** 92 min, +₹2.7 k per pack; the only design that passes at 2× heat
- **C · BMS observer for packs already built:** ₹0, safe at 44.5 °C, 277 min
- **Top risk:** the LFP cell is not validated; at 2× heat B misses the limit by 0.02 K and A becomes the pick
- **Next step:** a supervised LFP charge test (≤ 2S, fused) decides B or A

Data: Yao, Droese & Kowal (2026), TU Berlin, DOI 10.14279/depositonce-25110, CC BY 4.0. Costs and masses are generic class values. No design is claimed AIS-156 compliant. github.com/gitFa07/battery-energy-system ·

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